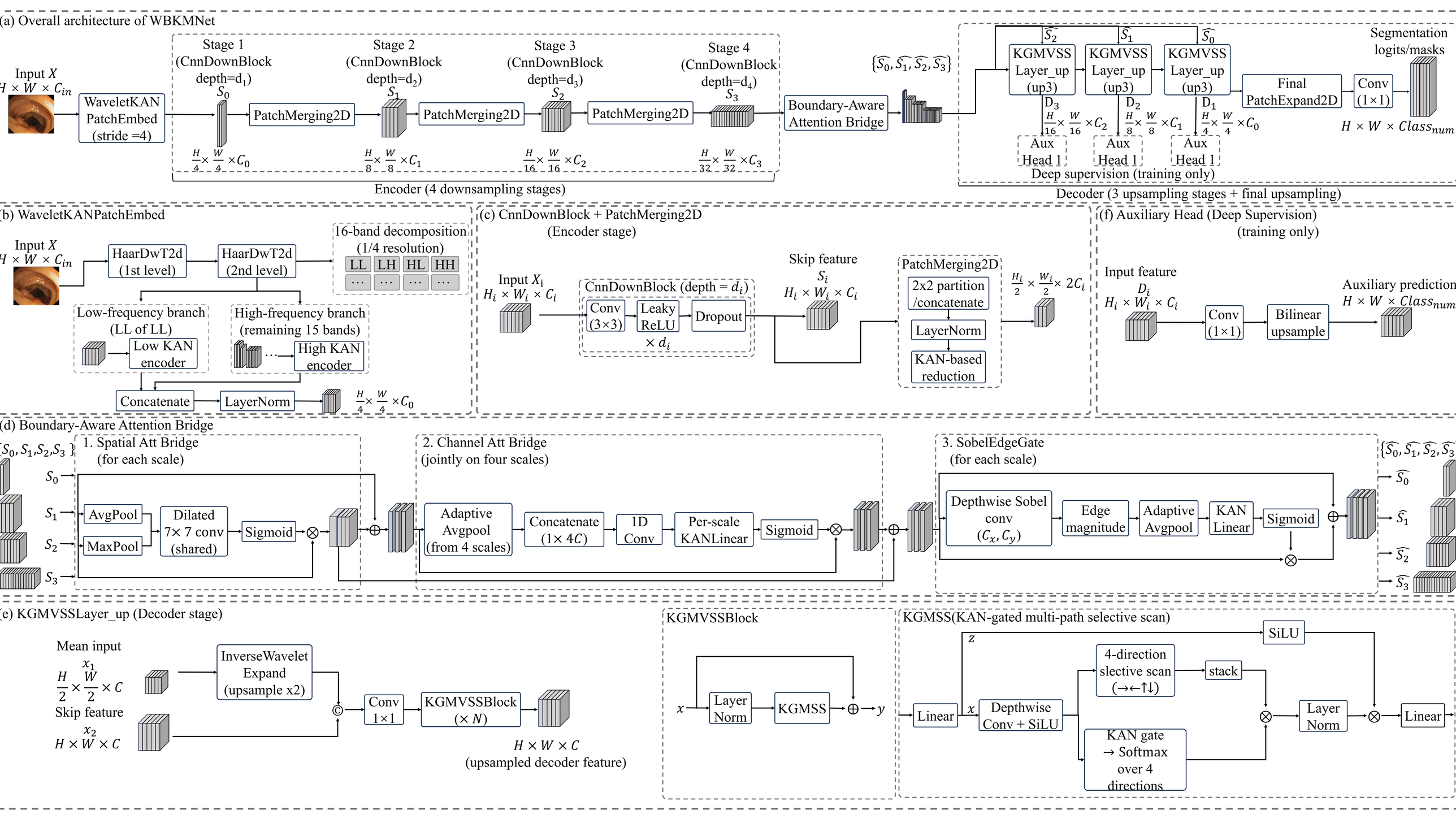

(a) Overall architecture of WBKMNet
Input X
$H \times W \times C_{in}$
WaveletKAN PatchEmbed (stride =4)
Stage 1 (CnnDownBlock depth=d_1)
Stage 2 (CnnDownBlock depth=d_2)
Stage 3 (CnnDownBlock depth=d_3)
Stage 4 (CnnDownBlock depth=d_4)
S_0
S_1
S_2
S_3
PatchMerging2D
PatchMerging2D
PatchMerging2D
$\frac{H}{4} \times \frac{W}{4} \times C_0$
$\frac{H}{8} \times \frac{W}{8} \times C_1$
$\frac{H}{16} \times \frac{W}{16} \times C_2$
$\frac{H}{32} \times \frac{W}{32} \times C_3$
Encoder (4 downsampling stages)
Boundary-Aware Attention Bridge
$\{\widehat{S_0}, \widehat{S_1}, \widehat{S_2}, \widehat{S_3}\}$
KGMVSS Layer_up (up3)
KGMVSS Layer_up (up3)
KGMVSS Layer_up (up3)
Final PatchExpand2D
Conv (1×1)
Segmentation logits/masks
$H \times W \times Class_{num}$
D_3 $\frac{H}{16} \times \frac{W}{16} \times C_2$
D_2 $\frac{H}{8} \times \frac{W}{8} \times C_1$
D_1 $\frac{H}{4} \times \frac{W}{4} \times C_0$
Aux Head 1
Aux Head 1
Aux Head 1
Deep supervision (training only)
Decoder (3 upsampling stages + final upsampling)
(b) WaveletKANPatchEmbed
Input X
$H \times W \times C_{in}$
HaarDwT2d (1st level)
HaarDwT2d (2nd level)
16-band decomposition (1/4 resolution)
LL LH HL HH
Low-frequency branch (LL of LL)
High-frequency branch (remaining 15 bands)
Low KAN encoder
High KAN encoder
Concatenate
LayerNorm
$\frac{H}{4} \times \frac{W}{4} \times C_0$
(c) CnnDownBlock + PatchMerging2D (Encoder stage)
Input X_i
$H_i \times W_i \times C_i$
CnnDownBlock (depth = d_i)
Conv (3×3)
Leaky ReLU
Dropout
$\times d_i$
Skip feature
S_i
$H_i \times W_i \times C_i$
PatchMerging2D
2x2 partition /concatenate
LayerNorm
KAN-based reduction
$\frac{H_i}{2} \times \frac{W_i}{2} \times 2C_i$
(f) Auxiliary Head (Deep Supervision) (training only)
Input feature
D_i
$H_i \times W_i \times C_i$
Conv (1×1)
Bilinear upsample
Auxiliary prediction
$H \times W \times Class_{num}$
(d) Boundary-Aware Attention Bridge
$\{S_0, S_1, S_2, S_3\}$
1. Spatial Att Bridge (for each scale)
AvgPool
MaxPool
Dilated 7× 7 conv (shared)
Sigmoid
2. Channel Att Bridge (jointly on four scales)
Adaptive Avgpool (from 4 scales)
Concatenate (1× 4C)
1D Conv
Per-scale KANLinear
Sigmoid
3. SobelEdgeGate (for each scale)
Depthwise Sobel conv (C_x, C_y)
Edge magnitude
Adaptive Avgpool
KAN Linear
Sigmoid
$\{\widehat{S_0}, \widehat{S_1}, \widehat{S_2}, \widehat{S_3}\}$
(e) KGMVSSLayer_up (Decoder stage)
Mean input
$\frac{H}{2} \times \frac{W}{2} \times C$
x_1
Skip feature
$H \times W \times C$
x_2
InverseWavelet Expand (upsample x2)
©
Conv 1×1
KGMVSSBlock (× N)
$H \times W \times C$
(upsampled decoder feature)
KGMVSSBlock
Layer Norm
KGMSS
x
y
KGMSS(KAN-gated multi-path selective scan)
Linear
Depthwise Conv + SiLU
4-direction slective scan (→←↑↓)
stack
KAN gate → Softmax over 4 directions
SiLU
Layer Norm
Linear